# TYPES OF STERILIZATION AND THEIR USES

sterilization processes are used every day around the world to eliminate hazardous biological agent and bacteria.This process is especially critical to the Medical,Pharmaceutical and Food industries for public safety and regulation compliance

## Steam (moist heat) Sterilization



Invented by Charles Chamberland in 1880

Steam Sterilization exposes each item to direct steam contact at the required temperature and pressure for the specified time.

Steam Sterilization is primarily used for heat stable materials such as:



Glassware

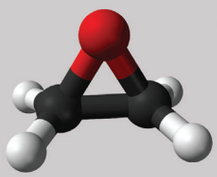


Surgical Instruments



Medical waste

## EtO/EO Sterilization



First used in the 1940s by the US military

EtO or EO sterilization is a method which utilizes Ethylene Oxide gas within a chamber to sterlize items or materials that cannot withstand the high temperatures or humidity that othersterilization methods employ.

Examples include devices or products including



Electrical components



Plastics



Gauze



Cardboard



Spices

## Depyrogenation /Dry Heat sterilization



Dry Heat Sterilization is one of the earliest forms of Sterilization practiced

Used on products that may be degraded when exposed to steam or moisture,but which can withstand high temperatures.

Examples of items sterilization by dry heat sterilization



Metal Surgical Instruments



Needle



Petroleum product



Glassware



Powders



Oils



TN_GOVT_XI_Micro_Biology_CH04.indd 50 16-02-2019 9.56.40 AM